Adaptive Exam-Prep Tutoring with a Transparent Learner Model

Research foundations and system design for a CBSE Class 10 Physics tutor

Research brief for hackathon problem statement #02
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Scope. This document consolidates the research behind each requirement of the problem statement (diagnostic test, concept graph, learner model, question generation with difficulty calibration, misconception detection from reasoning, spaced repetition, teacher dashboard), proposes a concrete system design, and defines an evaluation plan that works without real student data.

Evidence note. Peer-reviewed and preprint sources are cited throughout. Where a figure comes from a vendor blog or a non-peer-reviewed page, it is flagged in the text. Design recommendations are the authors' proposals, not findings from the cited studies.

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Abstract

General-purpose AI tutors can answer almost any question, but they do not maintain an evidence-based model of what an individual student understands. This paper reviews the research needed to build a tutor that does: one that runs a diagnostic test, represents a syllabus as a concept graph, updates mastery after every answer, targets questions at weak concepts with calibrated difficulty, detects misconceptions from the reasoning in a student's answer, schedules spaced review, and shows teachers where the whole class is stuck.

We find that (1) unguarded generative AI can improve practice performance while reducing later independent performance, so guardrails are a core design requirement; (2) for a system without a large training dataset, interpretable models (Bayesian Knowledge Tracing for mastery, an Elo-style rating for item difficulty) are better matched to the task than deep knowledge tracing; (3) detecting misconceptions in *correct* answers with flawed reasoning is a known blind spot of both classifiers and LLMs, which argues for pairing each question with a short explanation step and matching against a curated misconception library; and (4) modern spaced-repetition schedulers that model memory stability and retrievability outperform legacy fixed-ease schedulers in public benchmarks. We propose a system architecture for CBSE Class 10 Physics, an evaluation protocol based on simulated students with known ground truth, and a prioritised build plan.

Keywords: intelligent tutoring systems, knowledge tracing, learner modelling, misconception detection, spaced repetition, item response theory, Elo rating, teacher analytics.

1. Introduction and problem statement

The hackathon problem statement observes that generic AI tutors answer questions but do not track what a student actually understands. It requires teams to build seven components:

- a diagnostic test;
- a concept graph for one syllabus, such as Class 10 physics;
- a learner model that updates mastery for each concept after every answer;
- question generation targeted at weak concepts, with difficulty calibration;
- misconception detection from the reasoning in wrong answers, not just from the wrong final answer;
- spaced-repetition scheduling;
- a teacher dashboard showing class-wide gaps.

These components form a closed loop. The diagnostic seeds the learner model; the learner model selects the next question; each response updates the model and may reveal a misconception; the scheduler decides when a concept needs review; and the teacher dashboard aggregates the same state across a class. The quality of the whole system therefore depends less on any single component than on the learner model being accurate, inspectable, and wired into every other part.

> ***Central thesis.*** *The differentiator is not the chat interface. It is a learner model that demonstrably drives what the student sees next, and that can explain why. The website should make that visible.*

2. Why generic AI tutors fall short

2.1 Performance is not learning

A large field experiment in a high school (nearly 1,000 students, mathematics) compared three conditions: no AI, a standard chat interface built on GPT-4 ("GPT Base"), and a GPT-4 tutor with prompts designed to protect learning ("GPT Tutor") [1]. During AI-assisted practice, GPT Base students scored 48% better than the control group and GPT Tutor students 127% better. When access was removed for an exam, however, the GPT Base group scored 17% *worse* than control, while the GPT Tutor group performed about the same as control [1][2]. The authors attribute the harm to students using the model as a crutch; the safeguarded tutor, which gave hints and guidance rather than direct answers, largely avoided it [1][2].

> ***Design consequence.*** *Hint-first behaviour, withheld answers, and requests for the student's own reasoning are not cosmetic. They are the difference between a tool that raises practice scores and one that raises exam scores. The product should show its hint ladder and never default to revealing solutions.*

2.2 Availability is not engagement

Khan Academy reported that only about 15% of students with access to its Khanmigo AI tutor use it regularly, despite more than 108 million interactions since the 2023 rollout, and has said the tutor is being reworked after inconsistent early classroom results [3]. Its redesigned classroom platform adds a learner dashboard showing progress toward mastery and what to work on next [3]. This supports a product where the student's primary surface is a mastery-driven queue, with conversation as a supporting tool.

3. Learner modelling

3.1 Candidate approaches

A learner model estimates a student's knowledge state from their history of answers and predicts future performance. Three families dominate the literature.

- **Bayesian Knowledge Tracing (BKT).** A hidden Markov model per skill that treats knowledge as mastered or not mastered and updates the probability of mastery after each response. It has few parameters and is easy to interpret [4][5].
- **Logistic models** such as Performance Factors Analysis, which predict correctness from counts of past successes and failures and can be viewed as extensions of the Rasch model from item response theory [6].
- **Deep Knowledge Tracing (DKT)** and variants, which use recurrent or attention-based networks over the whole interaction sequence. Unlike BKT, DKT is a neural network with many free parameters, uses one network for all skills, and sees practice from multiple skills interleaved [5].

3.2 What the comparisons show

Khajah, Lindsey and Mozer showed that when BKT is extended to exploit recency, trial context, inter-skill similarity and individual ability, it reaches performance indistinguishable from DKT, and they argue that DKT's advantage does not come from discovering novel representations [4]. Their conclusion is that knowledge tracing may not require depth, and that shallow models give greater interpretability and explanatory power [4]. An empirical comparison across nine real-world datasets found that logistic regression with well-chosen features leads on moderate-sized datasets or those with many interactions per student, that DKT leads only on large datasets or where precise timing matters, and that classic Markov-process models such as BKT lag behind the others in raw predictive accuracy [7].

> ***Implication for a hackathon system.*** *There is no historical dataset to train a deep model, so DKT is the wrong tool. Raw predictive accuracy is also not the only objective: judges and teachers need to see why a concept is marked weak. We recommend BKT per concept as the explainable core, with prerequisite propagation, plus an Elo-style rating for question difficulty. A feature-based logistic model is a reasonable stretch goal once response data exists.*

3.3 Proposed mastery update (BKT)

For concept c, let L be the probability the student has mastered it, g the guess probability, s the slip probability and T the probability of learning on an opportunity. After an observed response:

$$P(L \mid \text{correct}) = L(1 - s) / [\, L(1 - s) + (1 - L)g \,]$$

$$P(L \mid \text{incorrect}) = L\, s / [\, L\, s + (1 - L)(1 - g) \,]$$

$$L_{\text{next}} = P(L \mid \text{response}) + (1 - P(L \mid \text{response}))\, T$$

Initial values for L, g, s and T can be set from the question type (for example, a four-option MCQ implies a guess floor near 0.25) and refined as data accumulates. To exploit the concept graph, evidence on a concept can be propagated to its prerequisites and dependents with damping, so that repeated failure on "series and parallel resistors" lowers confidence in "Ohm's law" even before Ohm's law is tested directly. This propagation rule is a design proposal of this paper rather than a published result, and should be evaluated in simulation (Section 11).

3.4 Difficulty calibration with Elo

The Elo rating system, designed for chess, has been adapted for education by treating each answer as a match between a student and a question [8][9]. After each answer, the student's ability rating and the item's difficulty rating are updated in opposite directions in proportion to the surprise of the outcome. In the Math Garden system, which collected more than 3.5 million arithmetic answers from 3,648 children over ten months, ability and item difficulty were updated after every answered item, response time was included in the scoring rule, and items were sampled with a mean success probability of about 0.75 so that tasks were challenging without being discouraging [9]. Pelánek argues that the approach is simple, robust and effective for adaptive educational systems [8].

$$P(\text{correct}) = 1 / (1 + e^{-(\theta - \delta)})$$

$$\theta \leftarrow \theta + K\,(X - P)$$

$$\delta \leftarrow \delta - K\,(X - P)$$

Here θ is student ability, δ is item difficulty, X is the observed response (1 or 0), and K controls update size. Known limitations apply: a critical analysis of Elo ratings in education notes that rating variance can inflate in certain adaptive scenarios, among other measurement shortcomings [10]. Practical mitigations are a decaying K (large for new items, small for established ones), clipping ratings to a range, and initialising difficulty from an LLM estimate or a teacher label rather than from zero.

4. Domain model: a concept graph for CBSE Class 10 physics

4.1 Syllabus facts

The CBSE Class 10 Science syllabus for 2026-27 applies to the 2027 board examinations [11]. Theory carries 80 marks, divided as Chemical Substances (25), World of Living (25), Natural Phenomena (12), Effects of Current (13) and Natural Resources (5), with 20 further marks for internal assessment [11][12]. The physics content sits mainly in Natural Phenomena, which covers Light (Reflection and Refraction) and The Human Eye and the Colourful World, and in Effects of Current, which covers Electricity and Magnetic Effects of Electric Current [11][13]. Third-party study sites differ on the exact mark split between individual physics chapters, so the final scope should be checked against the official syllabus document at cbseacademic.nic.in before questions are authored.

4.2 Recommended scope

The problem statement asks for one syllabus. We recommend a concept graph of roughly 40 to 60 concepts concentrated in Electricity and Light, because these chapters are numerical (supporting difficulty calibration), have well-documented misconceptions (supporting detection), and have natural prerequisite structure (supporting the graph). Magnetic Effects and the Human Eye can be added as stub nodes without full question banks.

4.3 Starter concept set (to be verified against NCERT)

ID	Concept	Prerequisites	Typical question types
E1	Electric charge and current	-	Definition, unit, direction of current
E2	Potential difference	E1	Definition, unit, voltmeter connection
E3	Circuit symbols and diagrams	E1	Draw and read circuits
E4	Ohm's law	E1, E2	Numericals, V-I graph interpretation
E5	Resistance and its factors	E4	Length, area, material; resistivity
E6	Resistors in series	E4, E5	Equivalent resistance, current and voltage sharing
E7	Resistors in parallel	E4, E5	Equivalent resistance, current division
E8	Heating effect of current	E4	Joule heating, applications
E9	Electric power and energy	E4, E8	P = VI, units, bills
L1	Laws of reflection	-	Ray diagrams, angle problems
L2	Image formation by spherical mirrors	L1	Concave and convex ray diagrams
L3	Mirror formula and magnification	L2	Sign convention numericals
L4	Laws of refraction	L1	Bending toward and away from the normal
L5	Refractive index	L4	Speed of light in media, numericals
L6	Image formation by lenses	L4	Convex and concave lens ray diagrams
L7	Lens formula, magnification and power	L5, L6	Numericals, dioptre

Table 1. Illustrative starter concept set. The ID scheme and prerequisite edges are a proposal and should be reviewed by a physics teacher.

4.4 Graph construction rules

- Draft nodes and edges with an LLM from the syllabus text, then have a human review every prerequisite edge. A wrong edge silently corrupts both question selection and propagation.
- Keep each node testable: a concept should be assessable by two to five distinct questions of different difficulty.
- Keep the graph a directed acyclic graph so that prerequisite propagation terminates and so that "learning path" views are well-defined.
- Attach to every node: a short description, one or more learning objectives, linked misconceptions, and links to questions.

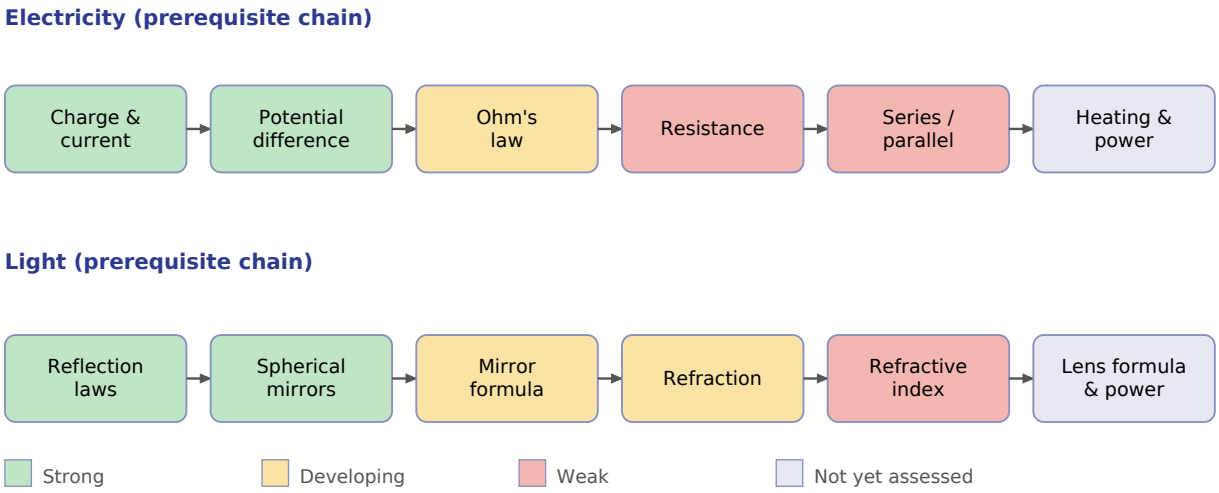


Figure 1. Illustrative prerequisite chains for Electricity and Light, coloured by an example mastery state. Left-to-right order is a simplified reading order; the full prerequisite edges are in Table 1. This is the central visual for both the student and teacher views.

5. Diagnostic test, question selection and generation

5.1 Diagnostic test

The diagnostic should establish an initial mastery estimate for every concept in as few questions as possible. A practical design is a short adaptive test of about 10 to 15 items. It starts from mid-graph concepts, moves toward prerequisites after errors and toward dependents after successes, and stops when uncertainty on most concepts falls below a threshold. Each item should include an "explain your answer" step so the misconception engine (Section 6) receives reasoning from the first question onward. The output is the initial graph colouring shown in Figure 1.

5.2 Choosing the next question

Item selection should balance three goals: practise the weakest concept that its prerequisites allow, choose difficulty so the predicted success probability is near a target (the Elo literature uses about 0.75 [9]), and reduce uncertainty about concepts the model is least sure of. A simple scoring function that combines these terms is sufficient, and the winning term should be shown to the student as a "why this question" line. Concepts that have been mastered enter the review scheduler (Section 7) rather than the practice queue.

5.3 LLM question generation with verification

LLMs can draft physics questions quickly, but they can produce wrong answer keys or distractors that are accidentally correct. We recommend generating questions from structured templates (concept, difficulty band, target misconception, numeric parameters) and then verifying them automatically: recompute numerical answers in code, check that exactly one option is correct, and confirm that each distractor corresponds to a named misconception or common calculation error. This follows the diagnostic-question principle that a question should not be answerable correctly while the student holds a key misconception [14]. Each generated question starts with an LLM- or teacher-estimated difficulty prior, which Elo then corrects from real responses.

6. Misconception detection from reasoning

This requirement is both the hardest and the most distinctive in the problem statement, which asks for detection from the reasoning in wrong answers rather than only from the final answer.

6.1 What the research shows

- **The correct-answer trap.** Students can reach correct answers through flawed reasoning, and a system that treats correctness as evidence of understanding will systematically under-detect misconceptions [14]. Existing benchmarks mostly predict which misconception explains a *wrong* answer; detecting a misconception in a correct answer is a distinct and less-studied problem [14].
- **Measured blind spot.** On 20,964 real student responses to 15 diagnostic mathematics questions from the Eedi platform, a fine-tuned BERT classifier detected only 57% of correct-answer misconceptions despite near-perfect classification of wrong answers [15]. Frontier LLMs were also reported to struggle with stepwise verification of student reasoning when the final answer is correct [14].
- **Label-space size matters.** Diagnostic accuracy of LLMs degrades sharply as the number of candidate misconceptions grows [14]. A small, curated library is therefore more reliable than open-ended labelling.
- **A workable pipeline.** A recent Eedi-led system generates a free-text misconception hypothesis with an LLM, retrieves the nearest tutor-defined labels in embedding space, then reranks candidates with a second model [16]. Framing the LLM as a hypothesis generator rather than a direct classifier gives an interpretable intermediate step [16].
- **Physics-specific evidence.** A classifier built on ModernBERT for motion-related misconceptions in introductory physics writing outperformed various prompting approaches on a small labelled set of 66 consented responses [17], which suggests that curated labels and light fine-tuning can beat generic prompting when data exists.
- **Keep a human in the loop.** Work on LLM analysis of student exam errors recommends using LLMs as a supplementary diagnostic tool rather than a substitute for instructor feedback [18].

6.2 Recommended design

- **Curated library.** Write 15 to 25 named misconceptions for the chosen scope, each with a description, linked concepts, example wrong beliefs in student language, and a remediation hint. Table 2 gives an illustrative starting set.
- **Distractor mapping.** Design MCQ distractors so each one corresponds to a library entry. A wrong choice then gives a strong prior on which misconception is active.
- **Explain-your-answer step.** After every answer (right or wrong), ask for one or two sentences of reasoning, or a choice among reasoning statements. This is how correct-answer misconceptions become visible.
- **Hypothesis, retrieve, rerank.** The LLM writes a one-sentence hypothesis of the student's belief; the system embeds it and matches against library entries; a second LLM pass confirms or rejects the top candidates and may return "none" or "uncertain".
- **Three-way output.** Report "misconception detected", "correct with sound reasoning" and "needs teacher review" rather than a binary flag, since a binary label can falsely accuse or falsely reassure [15].

- **Feedback loop.** Detected misconceptions lower the mastery estimate for linked concepts, trigger a targeted counter-example question, and appear in the teacher dashboard.

ID	Illustrative misconception	Linked concepts	Counter-example idea
M1	Current is used up by a bulb, so current is smaller after the bulb	E1, E8	Ammeter readings on both sides of a lamp in a series circuit
M2	A battery supplies the same current regardless of the circuit	E2, E4	Same cell, different resistors: compare currents
M3	Resistance of a conductor changes with the applied voltage or current	E4, E5	Same wire, V-I graph slope; vary length instead
M4	Adding a resistor in parallel increases total resistance	E7	Compute equivalent resistance for 1, 2, 3 identical branches
M5	In a series circuit every resistor has the same voltage across it	E6	Unequal resistors: measure voltage across each
M6	Covering half of a lens removes half of the image	L6	Ray diagram with partial obstruction: dimmer, complete image
M7	Light always bends toward the normal when entering any new medium	L4, L5	Glass-to-air and air-to-water cases
M8	Sign convention errors: treating distances on both sides of a mirror as positive	L3	Numerical with a convex mirror and virtual image

Table 2. Illustrative misconceptions of the kind reported in physics education research. These are proposals for the library and must be validated and refined by a physics teacher before use.

6.3 Evaluating without real misconceptions

Authentic student corpora with annotated cognitive mechanisms are scarce [19]. Recent work proposes generating synthetic reasoning traces that follow a specified misconception class, noting that producing an error consistent with a given failure mode is much harder than producing an arbitrary wrong answer [19]. The same idea lets us test our pipeline: generate synthetic student explanations from library entries, including correct-answer cases with flawed reasoning, then measure precision and recall of the detector. The limits of this approach are discussed in Section 13.

7. Spaced-repetition scheduling

7.1 From SM-2 to FSRS

SM-2, written by Piotr Woźniak in 1987, schedules each item using an interval and a single ease factor, and is known for "ease hell", where difficult items keep short intervals even after many successful reviews [20][21]. The Free Spaced Repetition Scheduler (FSRS), published in 2022 by Jarrett Ye and developed openly, models memory with three variables per item: difficulty, stability and retrievability [20][22]. The Anki project added FSRS as an alternative to its default scheduler [23].

Public benchmark figures from a 700-million-review Anki dataset, as summarised in a vendor guide, report log-loss of 0.354 for SM-2, 0.298 for FSRS-4.5 and 0.291 for FSRS-5, and retention-prediction RMSE of about 16.2%, 6.1% and 5.3% respectively; the same guide estimates roughly 25% fewer reviews at equal retention [21]. Because those sources are vendor-authored summaries of community benchmarks, the numbers should be treated as

indicative and should be re-checked against the open-spaced-repetition project's own repositories before being quoted publicly.

7.2 Proposed application to exam preparation

- **Schedule concepts, not individual questions.** The unit of review is a concept node with its own stability and retrievability, linked to the mastery model.
- **Exam-date-aware retention target.** Choose a desired retention that depends on the time to the board exam. One scheduler guide suggests about 90% for an exam two months away and a higher target for long-term mastery [22]. A visible "days to exam" control that reshuffles the review queue makes the scheduler tangible in a demo.
- **Link to the learner model.** A failed review lowers both retrievability and BKT mastery; a concept whose mastery drops below a threshold returns to the practice queue rather than the review queue.
- **Cold start.** Start from default FSRS parameters. Per-student parameter optimisation requires many reviews and is out of scope for the first version.

In the FSRS-4.5/5 family, the probability of recall after t days for a memory of stability S follows a power-law forgetting curve, and the next interval for a desired retention r can be solved in closed form. The constants below are the published defaults of that family and should be confirmed against the current reference implementation before use.

$$R(t, S) = (1 + F\, t / S)^{C}, \text{ with } F = 19/81 \text{ and } C = -0.5$$

$$\text{interval}(r) = (S / F)\,(r^{1/C} - 1)$$

8. Teacher dashboard

Existing products show that class-level analytics are expected. Khan Academy's classroom tooling includes a class snapshot for progress monitoring and instant grouping of students for small-group instruction [24], and an academic LLM-tutoring prototype describes a dashboard with visual analytics tracking class progress by topic [25]. A dashboard that only repeats averages by topic will not differentiate the product.

8.1 Recommended views

- **Class concept heatmap** overlaid on the prerequisite graph, so a teacher can see whether a failing topic is a root cause or a downstream symptom.
- **Misconception clusters.** For example, "14 of 32 students hold M1 (current is used up)", with representative anonymised student explanations and the confidence of the detection.
- **Suggested groups.** Students who share a weak concept or misconception, formed automatically for small-group instruction.
- **Action panel.** One click to generate a five-minute re-teach outline, three diagnostic questions that separate the misconception from careless errors, and a counter-example demonstration.
- **Teacher override.** A teacher can confirm, reject or relabel a detected misconception; these corrections become labelled data for improving the detector.
- **Individual drill-down** from a heatmap cell to a student's response history and review schedule, with the learner model's reasoning visible.

9. Competitive landscape

Summaries below draw on secondary sources (product pages and comparison articles) and should be refreshed before any public claim is made.

Product	Reported strengths	Gap relevant to this project
Khanmigo (Khan Academy)	Socratic questioning rather than direct answers [26]; teacher dashboard and grouping [24]; free classroom tools [3]	Chat-centred; low regular usage reported [3]; learner model not exposed to the student
ALEKS	Knowledge space theory for adaptive assessment and placement in STEM [27]	Tied to its publisher's content; proprietary models
Squirrel AI	Highly granular knowledge mapping and diagnostic assessment [27]	Limited presence outside its home market; not CBSE-specific
Century Tech and similar	Personalised pathways with teacher insights and diagnostics [26]	Generic curricula; misconception reasoning not the focus
This project	Transparent BKT + Elo model, misconception detection from reasoning, FSRS review linked to exam date, CBSE Class 10 physics focus	Narrow scope by design; needs validation with real students

Table 3. Competitive comparison.

10. System architecture

Figure 2 shows a modular architecture. The learner model is the shared state that every other component reads or writes. All LLM calls go through the server so keys and prompts are never exposed to the browser, and the orchestrator enforces the hint ladder so that no component can reveal an answer prematurely.

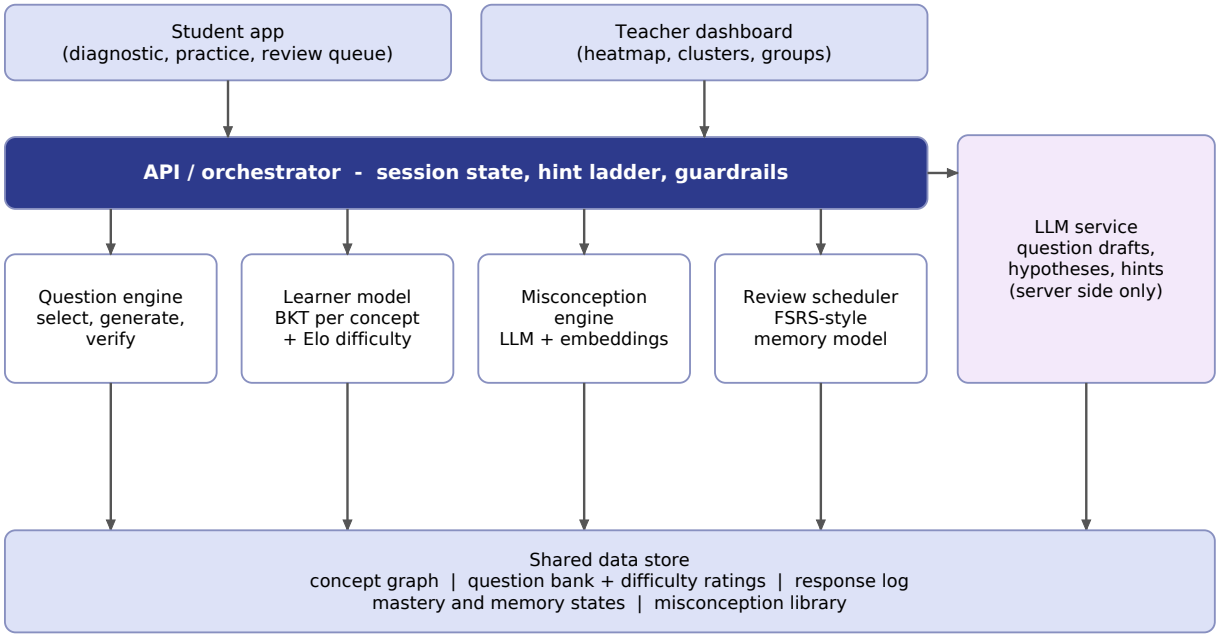


Figure 2. Proposed architecture. Arrows indicate primary data flow; the data store and learner model are shared by all engines.

10.1 Suggested implementation choices

- **Front end:** a React-based web app with a force-directed or layered graph view for the concept graph, a mastery heatmap component and a chart library for evaluation charts.
- **Back end:** a Python service for the learner model, scheduler and evaluation harness (convenient for numerical code), exposing a JSON API.
- **Storage:** a relational database for response logs, mastery and memory states; the concept graph and misconception library as versioned JSON files.
- **LLM use:** question drafting, hypothesis generation, hint writing and re-teach outlines, always followed by programmatic validation or a reviewer pass.
- **Embeddings:** a small sentence-embedding model is sufficient to match misconception hypotheses against a library of a few dozen labels [16].

11. Evaluation plan without real student data

A hackathon team will not have a cohort of real students. The next best evidence is a simulation in which the ground truth is known. This is also where the project can show more rigour than a typical prototype.

11.1 Simulated cohort

- Generate 30 to 200 synthetic students. Each has a hidden mastery value per concept, drawn so that mastery respects prerequisites, and one or two injected misconceptions from the library.
- Simulate answers with guess and slip noise. Wrong options chosen are determined by the active misconception, so distractor mapping is exercised.
- Generate free-text explanations from an LLM instructed to reason from the injected misconception, including cases where the student still reaches the correct answer. This follows the synthetic-trace approach in [19].

11.2 Metrics

Component	Metric	Baseline for comparison
Learner model	Error between estimated and true mastery over time; AUC and log-loss on next-answer prediction	Fixed-order questions; percent-correct per concept
Question selection	Questions needed to identify the weakest concepts; time to reach a target accuracy	Random order; syllabus order
Difficulty calibration	Rank correlation between learned item difficulty and simulated true difficulty	LLM difficulty prior alone
Misconception detection	Precision, recall and F1, reported separately for wrong-answer and correct-answer-with-flawed-reasoning cases	Keyword matching; direct LLM labelling without library
Scheduler	Simulated retention at the exam date versus number of reviews	Fixed intervals; SM-2-style scheduling
Dashboard	Accuracy of recovered misconception clusters and groups	Topic averages

Table 4. Proposed evaluation metrics.

11.3 Honest framing

Simulation shows that the mechanics work and that design choices matter; it does not show that students learn more. Every simulated number and every demo class must be labelled as simulated. If any real students can be recruited, even a small pilot of five to ten students with a teacher interviewing them about the diagnosis, that qualitative evidence is worth more than a larger synthetic study.

12. Website design recommendations

The website is the primary evidence that the system works. It should let a judge experience the loop in under two minutes and then find the depth underneath.

Page or section	Purpose	Key elements
Landing	State the problem and show the idea in seconds	One-line pitch; small live mastery map that changes when a visitor clicks a sample answer; link to demo
Diagnostic	Show adaptive assessment	10 to 15 adaptive questions with explain-your-answer; end with the concept graph reveal
Student view	Make the learner model visible	Mastery graph; today's review queue; "why this question next" panel; days-to-exam control
Practice session	Show guardrailed tutoring	Hint ladder before any reveal; misconception card with a counter-example; difficulty indicator
Teacher dashboard	Show class-wide gaps and actions	Heatmap; misconception clusters; suggested groups; re-teach generator; simulated-data label
How it works	Credibility	BKT, Elo and FSRS in plain language; evaluation charts; list of sources; limitations

Table 5. Proposed site structure.

- **Make state changes visible.** Animate the mastery colour change after each answer. The most persuasive moment in a demo is a student answering, the graph updating, and the next question visibly responding.
- **Explain every adaptive decision** in one short sentence ("Next: Ohm's law, because mastery is 0.42 and the prerequisite is secure").
- **Design for mobile.** Class 10 students are likely to use phones; graphs need tap-friendly nodes and readable labels at narrow widths.
- **Support bilingual use.** Offering Hindi and English question text is likely to matter for the target audience; this is a hypothesis to validate with teachers.
- **Accessibility.** Do not rely on colour alone for mastery; add labels or icons, and provide keyboard navigation.

13. Risks, ethics and limitations

Risk	Why it matters	Mitigation
LLM-written questions with wrong keys	Wrong questions corrupt both the student's learning and the difficulty ratings	Programmatic answer checks; teacher review of a sample; flag-this-question button
Answer leakage and over-reliance	Unguarded AI can reduce later independent performance [1]	Hint ladder enforced server-side; require a student attempt before any hint
Misconception false positives	Labelling a student wrongly can mislead remediation; hidden misconceptions are also under-detected [14][15]	Three-way output; confidence display; teacher override
Cold start and small data	Models trained on little data are unstable	Interpretable models with sensible priors; conservative update sizes
Elo rating variance	Ratings can inflate in adaptive settings [10]	Decaying K; clipped ratings; difficulty priors
Synthetic evaluation limits	Simulated students encode our own assumptions about how students err	Label as simulated; validate with a small real pilot
Children's data privacy	Class 10 students are typically minors	Collect minimal data; no real identifiers in the demo; confirm requirements under India's data-protection law (the DPDP Act, 2023, and its rules) before any real deployment
Curriculum drift	CBSE syllabi change between sessions [11]	Version the concept graph by session; verify against the official document

Table 6. Risk register.

Limitations of this review. Several cited findings come from preprints that have not completed peer review, and some figures come from vendor pages (flagged in the text). The proposed prerequisite propagation, exam-date-aware retention policy and three-way misconception output are design proposals, not validated results. The misconception library and concept graph in this paper are illustrative and need expert review. Findings on AI tutoring come mainly from mathematics and may not transfer exactly to physics.

14. Implementation priorities

Tier	Deliverable	Why in this order
1 (must)	Concept graph (about 40 nodes), question bank with verified answers, BKT mastery update, adaptive next-question selection, live graph view with "why this question"	This is the core loop that proves a learner model drives behaviour
2 (must)	Diagnostic flow; explain-your-answer step; misconception library and detector (hypothesis, retrieve, rerank); hint ladder	Covers the hardest and most distinctive requirement
3 (should)	Teacher dashboard on simulated class data: heatmap, clusters, suggested groups	Completes the problem statement's seventh requirement
4 (should)	FSRS-style review queue with days-to-exam control	Meets the scheduling requirement and adds a strong demo moment
5 (could)	Evaluation page with simulation charts; Hindi support; logistic-model baseline; teacher override feedback loop	Adds rigour and polish once the system works

Table 7. Prioritised build plan. Skip authentication, payments and multi-subject support.

15. Conclusion

The research points to a consistent design: use interpretable models where data is scarce, keep the LLM inside guardrails and behind validation, treat reasoning as first-class evidence alongside correctness, tie review scheduling to the learner model and the exam date, and expose the model's state to both students and teachers. A system built this way answers the problem statement's central criticism directly, because it can show, for any student at any moment, what the system believes they understand and why. The remaining work is to build the core loop, populate it with a carefully reviewed concept graph and misconception library, and evaluate it honestly.

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